

Complex Impedance and Phasor Relationships in AC Circuits

Gutierrez Kirby Benj, Chan Johrin Ashleigh, Tolentino Mark Samuel

March 2026

Introduction

In today's world no one can denied that electronics have become prevalent, and will continue its trends in prevalence within our time, so it's almost essential to have a basic understanding of its fundamentals. This paper aims to highlight not just the inner workings of circuits, but also the importance and elegant application of complex analysis.

In electrical engineering Alternating currents (AC) are essential in electronic systems, power transmission, communication network, and any modern devices that are used today, and it's safe to say that it plays a fundamental role in modern systems. Alternating currents (AC) circuits involve voltages and currents that vary periodically in relation to time, unlike direct current (DC) circuits, where voltage and current remain constant. This relation with time that AC circuits have introduced more complexity in circuit analysis, making traditional algebraic techniques insufficient for analyzing AC circuit systems.

Complex analysis provides an elegant way to solve this challenge, by representing voltages and currents as complex numbers. Using these differential equations that are traditionally used to describe circuit behavior can now be transformed into algebraic equations using complex analysis. Where traditionally quantities like voltage, current, and impedance needed to be solved individually it can now be expressed in magnitude and phases, greatly simplifying the problem.

When working with electricity, knowledge is invaluable. This task comes with its great deal of risks both during and after the work. An experienced Electrician would usually appear to do home repairs seamlessly, but that is not the case 100% of the time. Sometimes, the Job would be more complicated and would require a more thorough approach, and the preparation work requires doing difficult and lengthy calculations that involve a lot of calculus. However, this paper aims to show that with the applications of Complex Analysis, we can make reliable shortcuts in order to save precious time.

Methodology

I Approach

The methodology of this study uses a mathematical modeling approach to analyze how complex numbers can describe the behavior of electrical signals in alternating current (ac) circuits. This focuses on theoretical analysis using the existing electrical formulas and complex number representations instead of undergoing physical laboratory experiments.

Ac signals continuously change with time and follow the sinusoidal wave patterns. Analyzing these signals directly using trigonometric functions often leads to differential equations that are difficult to solve. Where sinusoidal signals are converted into complex numbers, phasor representation is used, which includes both phase information and magnitude to simplify the analysis.

According to **Euler's Formula**:

$$e^{j\theta} = \cos\theta + j\sin\theta$$

This identity allows sinusoidal signals to be represented as rotating vectors (phasors) in the Argand Plane. With this, derivatives of sinusoidal functions can be replaced by simple algebraic multiplication, making the circuit analysis easier.

II Representation of Electrical Quantities

In alternating current (AC) circuits, electrical quantities such as voltage and current change continuously with time. Meanwhile, the direct current (DC), where voltage remains constant, AC signals follow a repeating **sinusoidal waveform**. A typical AC voltage signal can be expressed as:

$$v(t) = V_0 \cos(\omega t + \theta)$$

where:

- V_0 represents the **maximum (peak) voltage**
- ω represents the **angular frequency of the signal**
- t represent **time**
- θ represents the **starting phase angle**

This equation shows that voltage changes with respect to time. Current in AC circuits behaves in a similar way, oscillating at the same frequency but possibly with a different phase showing that the peaks of the waves do not occur at the same time.

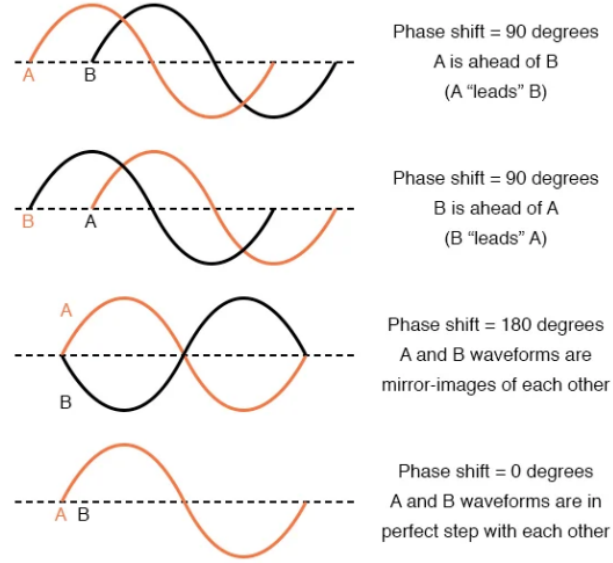


Figure 1: Sinusoidal Voltage and Current Waves

As shown in Figure 1 the relationship can be categorized into four states:

1. **90° Lead/Lag:** Caused by pure inductors or capacitors. This shows the lowest efficiency, where energy is stored rather than being consumed.
2. **180° Out of Phase:** where waveforms are mirror images, effectively canceling each other out.
3. **0° Phase Shift:** Where waveforms are in perfect step. This is the efficiency goal of complex number analysis: to bring the timing gap as close to zero as possible.

While the waveforms shown in Figure 1 accurately describe AC signals, solving problems using these time-dependent equations requires complex calculus. According to the subsection **Euler's Formula**, the voltage signal can be written as:

$$V = V_0 e^{j(\omega t + \theta)}$$

where:

- V_0 represents the **magnitude or peak value** of the signal.
- $e^{j(\omega t + \theta)}$ represents the **rotational component** in the Argand Plane.

III Static Phasor Modeling

To perform the algebraic solution mentioned in the **Approach**, we "freeze" these rotating vectors at $t = 0$. This allows us to represent the signal's magnitude and phase as the static coordinate.

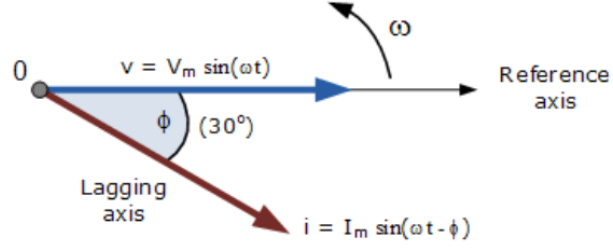


Figure 2: Phasor Representation on the Argand Plane.

A stationary vector representation where the angle corresponds to the wave's timing gap of 30° . By representing voltage and current in this way, the “timing gap” seen in Figure 2 is converted into a simple subtraction of angles ($\theta_v - \theta_i$), providing a direct measurement of the efficiency of the circuit.

IV The J Operator

To avoid confusion, the imaginary unit i , which commonly represents electric current in electrical engineering. The square root of negative one is represented by the imaginary unit:

$$j = \sqrt{-1}$$

In an AC circuit, multiplication by j corresponds to a 90-degree rotation in the complex plane. This property is useful because inductors and capacitors shift the phase of electric signals, and these shifts can be represented mathematically using the imaginary unit.

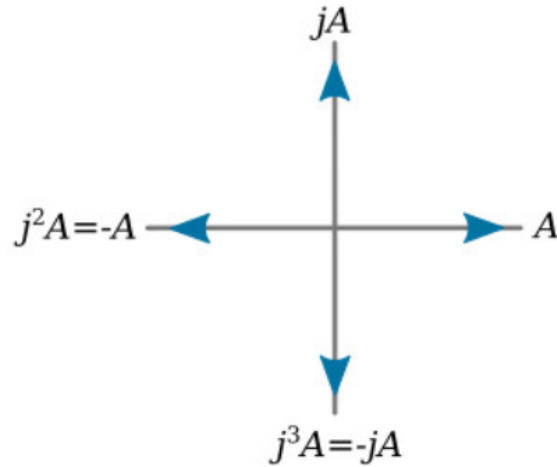


Figure 3: The J operator

If magnitude A is located on a set of coordinate axes as shown in the figure above, jA is 90° ccw, j^2A is 180° ccw, j^3 is 270° ccw, and j^4 is 360° ccw. This exponential progression is shown in Table 1 below.

Powers	Value	Rotation (CCW)
j^1	j	90°
j^2	-1	180°
j^3	$-j$	270°
j^4	1	360°

Table 1: Powers of the imaginary unit j

After the fourth power, the power repeats. This cyclic behavior is useful when performing algebraic operations with complex numbers in AC circuit analysis

V Polar and Rectangular Representation

While the j operator determines the rotation of a vector, complex numbers allow the location of any electrical quantity on a two-dimensional coordinate system known as the **Argand plane**. In this plane, the horizontal axis represents the **Real values** (resistance) and the vertical axis represents **Imaginary values** (reactance)

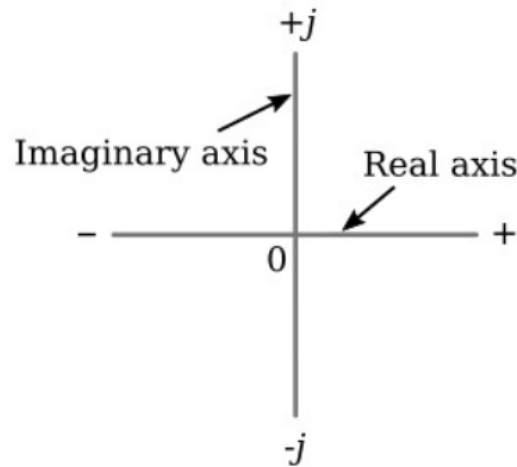


Figure 4: The Argand Plane

In **rectangular form**, a complex number is expressed as the sum of its horizontal and vertical components:

$$A = a + jb$$

Where:

- **a** is the **Real part**, representing the magnitude on the horizontal axis
- **b** is the **Imaginary part** representing the magnitude on the vertical axis

This form is useful when **adding** or **subtracting** phasors, such as calculating the total impedance of components in a series circuit.

In polar form, the same quantity can be defined by its Polar form, which describes the “resultant” vector:

$$A = r \angle \theta$$

or equivalently:

$$A = r(\cos\theta + j\sin\theta)$$

Where:

- **r** is the **magnitude**
- **θ** is the **Phase Angle**

The polar form is essential for understanding the **efficiency** of a circuit, as the angle directly represents the phase shift or timing gap between voltage and current. It is preferred for **multiplying** or **dividing** phasors.

To Transition between two representations, trigonometric relationships are applied as shown at the table below:

Form	To	Formula
Rectangular ($a + jb$)	Magnitude(r)	$r = \sqrt{a^2 + b^2}$
Rectangular ($a + jb$)	Angle(θ)	$\theta = \arctan \frac{b}{a}$
Polar ($r \angle \theta$)	Real(a)	$a = r \cos \theta$
Polar ($r \angle \theta$)	Imaginary(b)	$b = r \sin \theta$

Table 2: Conversion formula of a Complex Number

VI Complex Impedance Model

In direct current (DC) circuits, resistance is the primary quantity that opposes the flow of electric current. Meanwhile, in alternating current (AC) circuits, additional components such as **inductors** and **capacitors** are included, which influence both the magnitude and the timing of electrical signals. To represent the effects, electrical engineers used a quantity called **impedance**.

Impedance is the total opposition to current flow in an AC circuit and is expressed as a **complex number**. This representation allows both resistance and reactance to be analyzed simultaneously and written as:

$$Z = R + jX$$

Where:

- **R** represents **resistance**, which dissipates electrical energy as heat
- **X** represents **reactance**, which stores and releases energy through an electric or magnetic field
- **j** represents the **imaginary unit**

This complex representation allows to apply a generalized form of Ohm's Law:

$$V = IZ$$

Where:

- **V** complex voltage
- **I** is the current
- **Z** is the complex impedance

Unlike direct current circuits, where the resistance is constant, the impedance of inductors and capacitors depends on the **frequency of the signal**. The impedance of the three fundamental circuit components is summarized in the following table:

Components	Impedance Formula	Electrical Effect
Resistors	$Z_R = R$	Voltage and current remain in the phase
Inductors	$Z_L = j\omega L$	Current lag voltage
Capacitors	$Z_C = \frac{1}{j\omega C} = \frac{-j}{\omega C}$	Current lead voltage

Table 3: Impedance of Basic AC circuit Components

Where:

- **L** represents **inductance**
- **C** represents **capacitance**
- ω represents the **angular frequency of the AC signal**

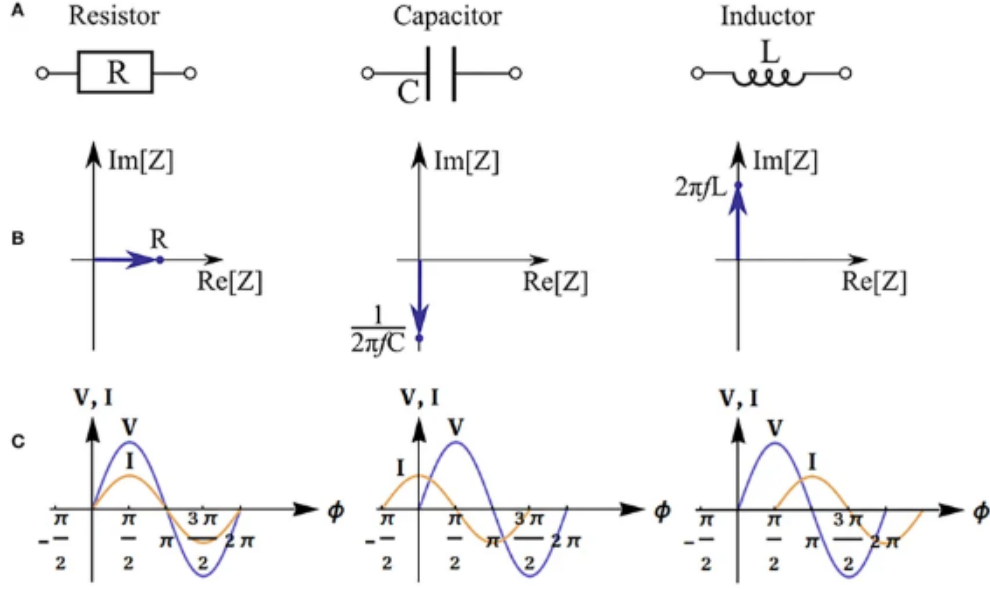


Figure 5: Impedance Behavior of Resistors, Inductors, and Capacitors

The Figure 5 shows the phase relationship between voltage and current for the three fundamental Alternating Circuit components. In a resistor, voltage and current remain aligned, showing that they are in **phase** and reach their maximum and minimum values **simultaneously**. In an inductor, the current lags behind the voltage by 90° , while in a capacitor, the current **leads** the voltage by 90° .

VII Circuit Modeling Procedure

To analyze the behavior and efficiency of an Alternating Circuits using the complex impedance model, the following step-by-step procedure is applied.

- **Step 1: Define Circuit Parameters**

Identify the electrical parameters of the circuit, including the peak voltage V_0 , operating frequency f , and the values of the circuit components such as resistance R , inductance L , and capacitance C .

- **Step 2: Calculate Angular Frequency**

Identify the angular frequency using the following relationship:

$$\omega = 2\pi f$$

- **Step 3: Map Components to Complex Impedance**

Convert each physical component into a corresponding complex impedance representation using the formulas introduced in the **Complex Impedance Model** subsection:

- $Z_R = R$
- $Z_L = j\omega L$
- $Z_C = \frac{1}{j\omega C}$

This converts the physical circuit into a mathematical model located on the **Argand Plane**.

- **Step 4: Determine the Total Impedance**

For a **series circuit**, the total Impedance Z_{total} is obtained by adding the individual impedance components in rectangular form:

$$Z_{total} = Z_R + Z_L - Z_C$$

- **Step 5: Apply Generalized Ohm's Law**

Solve for the **current phasor** I by dividing the voltage phasor by the total impedance:

$$I = \frac{V}{Z_{total}}$$

- **Step 6: Convert Results to Polar Form**

Transform the final rectangular result into **polar form** ($r\angle\theta$). This is to find the resulting magnitude and the specific **phase angle**.

- **Step 7: Evaluate Circuit Efficiency**

Analyze the resulting phase angle θ to determine the "timing gap" between voltage and current. A smaller angle indicates a more efficient circuit with a **higher power factor**.

Result

Case 1: Purely Capacitive AC Circuit

- **Circuit Parameters:**

To demonstrate the application of the complex impedance model, a **purely capacitive AC circuit** was analyzed. The parameters used in the analysis are presented in the Table below:

Parameters	Symbol	Value
Voltage Phasor	V	$150\angle 40^\circ$
Frequency	f	$15kHz$
Capacitance	Z_C	$0.06\mu F$

Table 4: Electrical Parameters of the capacitive circuit

- **Angular Frequency:**

The angular frequency was calculated using:

$$\omega = 2\pi f$$

$$\omega = 2\pi(15000)$$

$$\omega = 9.42 \times 10^4 \text{ rad/s}$$

- **Capacitive Impedance:**

The impedance of the capacitor is given by:

$$Z_C = \frac{1}{j\omega C}$$

Substitute the given values:

$$Z_C = \frac{1}{j(9.42 \times 10^4)(0.06 \times 10^{-6})}$$

$$Z_C = -j177\Omega$$

In polar form:

$$Z_C = 177\angle -\frac{\pi}{2}\Omega$$

- **Current Phasor:**

The current was solved using the generalized form of Ohm's Law:

$$I = \frac{V}{Z}$$

Substitute the given values:

$$I = \frac{150\angle 40^\circ}{177\angle -90^\circ} = 0.847\angle 130^\circ$$

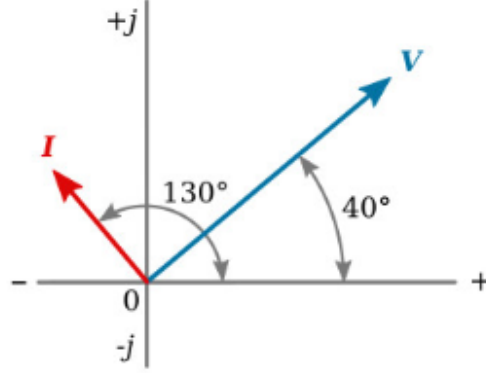


Figure 6: The Phasor Diagram for a Capacitive Circuit

The Figure 6 shows the voltage phasor V at 40° and the current phasor I at 130° .

• **Data Analysis:**

Parameters	Magnitude	Phase Angle
Voltage V	150V	40°
Current I	0.847A	130°
Timing Gap ϕ	—	$+90^\circ$ (Lead)

Table 5: Summary of Phasor Result.

As shown in Table 5, the voltage V was established at a reference angle of 90° . The resulting current phasor I was calculated to be 0.847A at an angle of 130° . By subtracting the phase angle ($\theta_I - \theta_V$), we find:

$$130^\circ - 40^\circ = +90^\circ$$

To solve for the % efficiency:

$$\cos(\theta) \times 100 = \cos(90^\circ) \times 100 = 0\%$$

This $+90^\circ$ **timing gap (or phase shift)** proves that the current **leads** the voltage with 0% **efficiency**. This confirms the “ICE” rule (In a capacitor, Current leads).

Case 2: Mixel RL Circuit

In this case, the circuit has both Resistance Z_R and Inductance Z_L .

• **Circuit Parameters:**

Given an example of a given parameters of:

Parameters	Symbol	Value
Voltage Phasor	V	$100\angle 0^\circ V$
Frequency	f	$60Hz$
Resistance	Z_R	8.66Ω
Inductance	Z_L	$13.26mH$

Table 6: Electrical Parameters of the Mixed RL circuit

- **Angular Frequency:**

The angular frequency was calculated using:

$$\omega = 2\pi f$$

$$\omega = 2\pi(60)$$

$$\omega = 376.99rad/s$$

- **Inductive Impedance:**

The impedance of the capacitor is given by:

$$Z_L = \omega L$$

Substitute the given values:

$$Z_L = (376.99rad/s)(0.01326mH)$$

$$Z_L = 5\Omega$$

- **Total Impedance:**

Combining the Real resistance and Imaginary reactance into rectangular form:

$$Z_{total} = Z_R + Z_L = 8.66 + j5\Omega$$

- **Convert to Polar Form:**

To find the timing gap, the Total Impedance is converted coordinates into **a magnitude and angle**:

- **Magnitude (Z):** $\sqrt{8.66^2 + 5^2} = 10\Omega$

- **Phase Angle(θ) :** $\arctan(\frac{5}{8.66}) = 30^\circ$

– **Polar Form:** $Z = 10\angle 30^\circ$

- **Current Phasor:**

The current was solved using the generalized form of Ohm's Law:

$$I = \frac{V}{Z}$$

Substitute the given values:

$$I = \frac{100\angle 0^\circ}{10\angle 30^\circ} = 10\angle -30^\circ$$

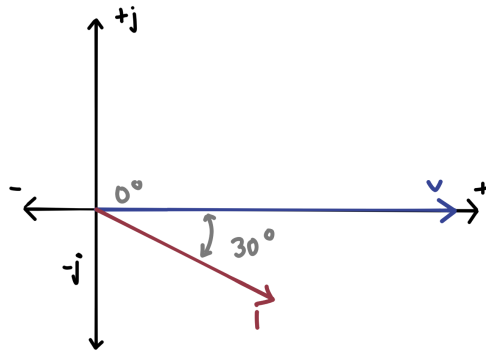


Figure 7: The Phasor Diagram for a Mixed RL Circuit

The diagram shows the voltage phasor V at 0° and the current phasor I at -30°

- **Data Analysis:**

Parameters	Magnitude	Phase Angle
Voltage V	100V	0°
Current I	10A	-30°
Timing Gap ϕ	—	-30° (Lead)

Table 7: Summary of Phasor Result.

As shown in Table 8, the voltage V was established at a reference angle of 0. the resulting current phasor I was calculated to be 10A at an angle of -30 . By subtracting the phase angle ($\theta_I - \theta_V$), we find:

$$-30 - 0 = -30$$

To solve for the % efficiency:

$$\cos(\theta)100 = \cos(-30) \times 100 = 86.60\%$$

This **−30 timing gap (or phase shift)** proves that the current **Lags** the voltage with 86.60% efficiency, means that the Circuit has a Good Efficiency **with a minor waste**.

Discussion

By using ideas and methods from Complex Analysis a “shortcut” has been made letting the process of finding “circuit efficiency” become significantly shorter than it usually would be. This is achieved through using Euler’s Formula, where rotating vectors can be shown and represented as complex numbers. Through this, the presented j operator represents shifts of 90 degrees that are created by different physical components of a circuit. The inclusion of Resistance, inductance, and Capacitance also have their own individual solutions but with this, we can already include them alongside the Voltage Phasor to be our parameters.

Parameters	Symbol	Value
Voltage Phasor	V	$100\angle 0^\circ V$
Frequency	f	$60Hz$
Resistance	Z_R	8.66Ω
Inductance	Z_L	$13.26mH$

Table 8: Electrical Parameters of the Mixed RL circuit

Instead of doing all the formulas for each individual parameter we can also simplify them to this equation:

$$Z_{Total} = Z_R + Z_L - Z_C$$

In which the following adds direction to our graph on the Argand plane, each value representing:

- Z_R = Resistor: Always pointing **Right**: 0 this is **Pure Real**
- Z_L = Inductor: Always pointing **UP**: +90 this is **+J**
- Z_C = Resistor: Always pointing **Left**: −90 this is **-J**

After boiling down a lengthy solution to simpler versions, such as the j operator, all that is left is to apply simple algebra alongside the utilization of a generalized version of Ohm’s Law to solve for current or voltage. Now that most of the process has been performed, the results need to be converted to polar form so that the magnitude and phase angle can be found, and the purpose of finding the phase angle is because it is the one that represents the timing gap, which is the main value needed in determining or evaluating the circuit efficiency.

Conclusion

Complex Analysis proves to be an effective tool to shorten work that includes lengthy calculations by simplifying its components, given the right circumstances. This paper shows that exact feat through the use of the created J operator to aid in electrical component work.

A lot of risk exists in Electrical work; a simple mistake can lead to consequences ranging from inconveniences to life-threatening accidents. Studies such as these can help improve both work efficiency and risk reduction, which is why this study proved to be very insightful in its essence.

Complex Analysis is only one part in the whole field of mathematics and already has proved to be valuable. Proper integration and application of different areas may be able to achieve greater results and lead to discoveries and innovation, which Academic Researchers should always aim to do

References

- Centennial College Library. (n.d.). *Apa citation guide*. <https://libraryguides.centennialcollege.ca/c.php?g=717490&p=5120896>
- EC Studio Systems. (n.d.). *Complex numbers in electronics*. <https://ecstudiosystems.com/discover/textbooks/basic-electronics/ac-circuits/complex-numbers-in-electronics/>
- Electronics Tutorials. (n.d.). *Phasors*. <https://www.electronics-tutorials.ws/accircuits/phasors.html>
- GeeksforGeeks. (2023). *Impedance in ac circuits*. <https://www.geeksforgeeks.org/electrical-engineering/impedance-in-ac-circuits/>
- Gerasimenko, T., Nikulin, S., Zakharova, G., & Tonevitsky, A. (2020). *Impedance of basic equivalent circuit components: Resistor, capacitor, and inductor* [Figure from: Impedance spectroscopy as a tool for monitoring performance in 3D models of epithelial tissues]. https://www.researchgate.net/figure/Impedance-of-basic-equivalent-circuit-components-resistor-capacitor-and-inductor_fig2_338794880
- Loinaz, W. (2003). Ac circuits [Physics 34 Laboratory Handout]. <https://waloinaz.people.amherst.edu/P34/P34s03labs/P34AC-Circuits.pdf>
- Nave, C. R. (n.d.). *Complex impedance*. <http://hyperphysics.phy-astr.gsu.edu/hbase/electric/impcom.html>
- OpenStax. (2016). *Precalculus 1e*. [https://math.libretexts.org/Bookshelves/Precalculus/Precalculus_1e_\(OpenStax\)/08%3A_Further_Applications_of_Trigonometry/8.05%3A_Polar_Form_of_Complex_Numbers](https://math.libretexts.org/Bookshelves/Precalculus/Precalculus_1e_(OpenStax)/08%3A_Further_Applications_of_Trigonometry/8.05%3A_Polar_Form_of_Complex_Numbers)